University of Asia Pacific

Department of Computer Science and Engineering

Course Code : CSE 450 (VLSI Lab)

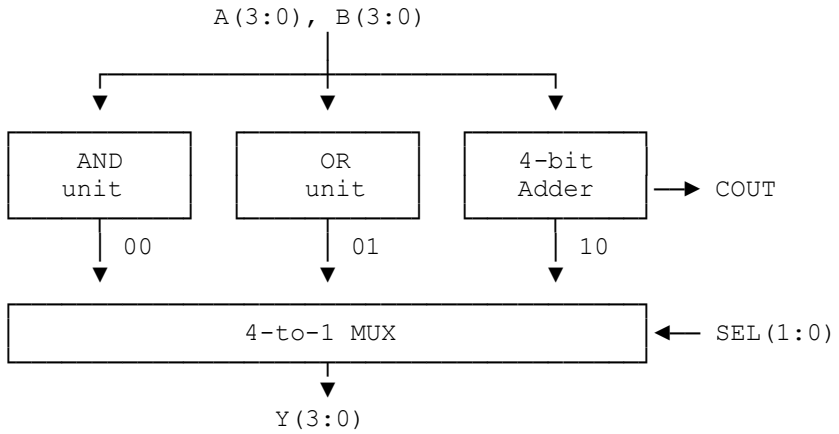
Mid-Term Lab Examination | Group B | Spring 2026

Duration	1 hr	**Total Marks**	20
Student Name	_____	**ID**	_____

Mid-Term Task: 4-bit ALU

Objective. Design and implement a 4-bit Arithmetic Logic Unit (ALU) in VHDL that supports exactly three operations: AND, OR,, and ADD. The ALU must be built from separate, reusable modules (logic units, a 4-bit adder, and a multiplexer) and combined in a top-level entity.

System Architecture



Functional requirement. The 2-bit select input SEL chooses which result is passed to the output Y. The AND and OR units operate bitwise on A and B, while the adder produces a 4-bit sum together with a carry-out. The ALU is purely combinational: Y and COUT must change whenever A, B, or SEL change. COUT must be 0 for every operation other than ADD.

Required Operation

SEL(1:0)	Operation	Y(3:0)	COUT
00	AND	Y ← A AND B	0
01	OR	Y ← A OR B	0
10	ADD	Y ← (A + B) mod 16	Carry-out of the 4-bit addition
11	Unused	Y ← 0000	0

Required Test Sequence

Your testbench must apply the following sequence. Allow each input combination to settle (for example, wait for 10 ns) before checking the outputs.

Step	SEL	A	B	Operation	Expected Y	COUT	Explanation
1	00	1100	1010	A AND B	1000	0	Bitwise AND
2	00	1111	0101	A AND B	0101	0	AND with all-ones passes B
3	01	1100	0011	A OR B	1111	0	Bitwise OR
4	01	1001	0100	A OR B	1101	0	Bitwise OR
5	10	0011	0101	3 + 5	1000	0	Simple addition
6	10	1111	0001	15 + 1	0000*	1	4-bit overflow
7	10	1010	0101	10 + 5	1111	0	Maximum 4-bit result
8	11	1010	0101	Unused	0000	0	Invalid select code

*** Note:** For Step 6, the mathematical result is 16, but a 4-bit output can represent only 0–15. Therefore, Y = 0000 and the discarded carry appears on COUT = 1.

Implementation Requirements

- Use VHDL and IEEE.STD_LOGIC_1164.ALL.
- Ports: A, B : in STD_LOGIC_VECTOR(3 downto 0); SEL : in STD_LOGIC_VECTOR(1 downto 0); Y : out STD_LOGIC_VECTOR(3 downto 0); COUT : out STD_LOGIC.
- Build the 4-bit adder structurally from 1-bit full adders (ripple-carry). Do not use the "+" operator or an integer/numeric_std based adder as a replacement.
- The design must be modular: separate entities for the AND unit, OR unit, full adder, 4-bit adder, and 4-to-1 multiplexer, connected by a structural top-level ALU.
- The ALU must be combinational; no clock or process-based registers are allowed.
- SEL = "11" must produce Y = "0000" and COUT = '0'.
- The submitted testbench must apply all eight steps of the required test sequence and use ASSERT statements to compare the actual output with the expected output.

Submission

While submitting to the classroom prepare the report from the third page of this template including the current template in the first and second page. From the third page write your report including but not limited to the following points.

- Attach your entity code and testbench code
- Attach your output visuals (waveform showing all eight test steps)
- Challenges faced during the experiment
- Conclusion